C26 Carbonyl Compounds – Knowledge Organiser

AQA A-level Chemistry • Set 1: Aldehydes and Ketones **Paper 2**

#	Question	Answer
1	Reducing agent for aldehydes and ketones	$NaBH_4$
2	Product of reducing an aldehyde	Primary alcohol
3	Product of reducing a ketone	Secondary alcohol
4	Symbol for a reducing agent in equations	[H]
5	Reagents to form a hydroxynitrile	KCN, then dilute acid
6	Product of ethanal with KCN/H^+	2-hydroxypropanenitrile
7	Why KCN is used rather than HCN	HCN is a toxic gas
8	Mechanism type for CN^- or $NaBH_4$ with C=O	Nucleophilic addition
9	Product of KCN with an aldehyde (optical activity)	Racemic mixture
10	Hazard of KCN	Toxic

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AQA A-level Chemistry • Set 2: Carboxylic Acids and Esters **Paper 2**

#	Question	Answer
1	Products of a carboxylic acid with a carbonate	Salt + CO_2 + H_2O
2	Reagents to make an ester	Carboxylic acid + alcohol, conc H_2SO_4 catalyst
3	Ester suffix	-oate
4	Ester from ethanol and ethanoic acid	Ethyl ethanoate
5	Uses of esters	Perfumes, flavourings, solvents, plasticisers
6	Products of acid hydrolysis of an ester	Carboxylic acid + alcohol
7	Products of alkaline hydrolysis of an ester	Carboxylate salt + alcohol
8	Fats and oils	Esters of glycerol (propane-1,2,3-triol)
9	Soap	Sodium or potassium salt of a long-chain fatty acid
10	Biodiesel	Methyl esters of long-chain fatty acids

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AQA A-level Chemistry • Set 3: Acylation **Paper 2**

#	Question	Answer
1	Acyl chloride functional group	–COCl
2	Acid anhydride general formula	$(RCO)_2O$
3	Acyl chloride + water	Carboxylic acid + HCl
4	Acyl chloride + alcohol	Ester + HCl
5	Acyl chloride + ammonia	Primary amide + HCl
6	Acyl chloride + primary amine	*N*-substituted amide + HCl
7	Mechanism type for acylation	Nucleophilic addition–elimination
8	Observation when an acyl chloride reacts with water	Steamy fumes (HCl)
9	Why anhydrides are used industrially	Cheaper, less corrosive, less violent, no HCl
10	Reagents to make aspirin	2-hydroxybenzoic acid + ethanoic anhydride